Question 1

Aptitude | Logical Reasoning 2

There are four pens: a red pen, a black pen, a green pen and a blue pen. The pens need to be kept in a shelf from left to right. The red pen should be on the left. The black pen should be kept just after the red pen and the blue pen should be on the right. The green pen should be on the left of the blue pen but on the right of the black pen. Which of the following pen is kept on the left of the black pen?

A. Red

B. Green

C. Blue

D. Cannot be determined

Independently solved answer: A

Question 2

Aptitude | Puzzles 2

Pointing to a photograph Rita, said “he is the son-in-law of the grand mother of my father’s father’s only son”. How is Rita’s father related to the wife of the man in photograph?

A. Son-in-law

B. Brother-in-law

C. Nephew

D. Brother

Independently solved answer: Ambiguous; C likely intended

Question 3

Aptitude | Logical Reasoning 2

For runners (R1, R2, R3, and R4) completed a marathon. R1 finished before R4 but after R2. R2 finished after R3. Who finished the marathon first?

A. R1

B. R2

C. R3

D. R4

Independently solved answer: C

Question 4

Aptitude | Logical Reasoning 2

Five teachers I, J, K, L and M teach in four different schools. M and K do not teach in DPS, L teaches in St. Paul's and I teaches in St. Xavier's. If two teachers out of the five teach in Model school, which school does M teach in?

A. St. Paul's

B. St. Xavier's

C. Model School

D. Cannot be determined

Independently solved answer: C (assuming all four stated schools are represented)

Question 5

Aptitude | Puzzles 2

How many such pairs of letters are there in the word **‘PRECISE’** which has as many letter between them (from both the sides) as in the alphabets from A to Z?

A. 0

B. 1

C. 2

D. 3

Independently solved answer: A

Question 6

Aptitude | Logical Reasoning 2

Direction: Study the information given below and answer the question that follows.

Five friends - Rohan, Sia, Tanu, Varun, and Zoya - save money in their piggy banks.

- Together, they save ₹1,150.

- Rohan saves ₹150 more than Sia.

- Tanu saves twice as much as Sia.

- Varun saves the same amount as Rohan.

- Zoya saves ₹50 less than Sia.

How much does Tanu save?

A. ₹250

B. ₹280

C. ₹300

D. ₹320

Independently solved answer: C

Question 7

Aptitude | Puzzles 1

If today is Wednesday, _____ will be the day of the week after 15 days.

Independently solved answer: Thursday

Question 8

Aptitude | Puzzles 1

Direction: Study the following data carefully and answer the questions accordingly.
A number arrangement machine, when given an input line of numbers, rearranges them following a particular rule in each step. The following is an illustration of input and rearrangement:

Input	531842	764125	902317	648531	287964	413256
Step I	8	12	3	15	21	9
Step II	15	13	19	12	15	12
Step III	289	313	370	369	666	225
Step IV	64		353		1	
Step V	10		11		1	

Step V is the last step of the rearrangement. As per the rules followed in the above steps, find out the output for the below input and answer the question followed.

Input	825431	370962	641528	954210	718346	260871

Step V of the above input is _____ .

Independently solved answer: 3, 6, 18

Question 9

Aptitude | Quant Math Ability 2

A U.S. streaming service offers three plans: Basic at $15, Standard at $25, and Pro at $45 per month. There are 600 subscribers in total, and the average monthly revenue per subscriber is $26. Also, the number of Pro subscribers is 60 less than the number of Basic subscribers. How many Standard subscribers are there?

A.
280

B.
340

C.
320

D.
300

Independently solved answer: D

Question 10

Aptitude | Quant Math Ability 2

Three men P, Q and R working individually take 12 days, 15 days and 20 days respectively to construct a wall. Find the minimum time required to construct a wall if not more than 2 men are working together on a day and no same pair of men work on two consecutive days.

A. 6.5 days

B. 7 days

C. 7.5 days

D. 8 days

Independently solved answer: B

Question 11

Aptitude | Quant Math Ability 2

A management club has 5 distinct technical activities and 4 distinct strategy activities. For a short event, exactly 2 technical activities and exactly 1 strategy activity must be selected. The selected 3 activities will be conducted in three distinct slots: Slot 1, Slot 2, and Slot 3. The strategy activity must not be conducted in Slot 1.
In how many ways can the event schedule be prepared?

A.
120

B.
140

C.
160

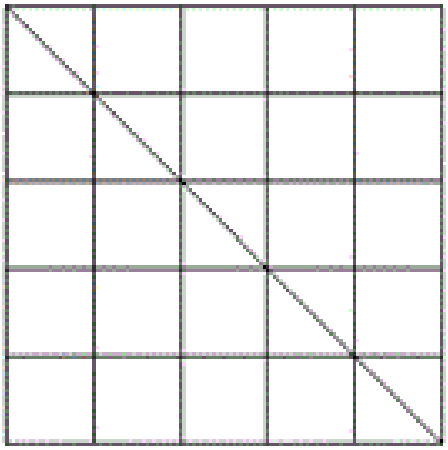
D.
180

Independently solved answer: C

Question 12

Aptitude | Puzzles 2

Use the figure given below to answer the question given below:



What is the number of triangles in the figure?

A. 30

B. 27

C. 20

D. 15

Independently solved answer: A

Question 13

Aptitude | Quant Math Ability 1

15 books are placed on a table. The probability that three of the books will always be together is x/y. The value of x is 1 and the value of y is _____ .

Independently solved answer: 35

Question 14

Aptitude | Logical Reasoning 1

In a queue of more than 10 doctors, Dr. A is 9th from the left end and Dr. B is 3th from the right end, there are three doctors between Dr. A and Dr. B. A total of _____ doctors are there in the queue.

Independently solved answer: 15

Question 15

Aptitude | Logical Reasoning 2

Direction: Study the information given below and answer the question that follows.

Seven football players namely Gaurav, Harsh, Kunal, Lalit, Manish, Naman and Ojha are to be selected for one of the college teams, the Phoenix and the Devils.

- If Gaurav is selected for the Phoenix, then Harsh will be selected for the Devils
- If Kunal is selected for the Phoenix, then Lalit and Manish will be selected for the Devils.
- Naman is selected on a different team than Ojha.
- Manish is selected on a different team than Gaurav.
- If Ojha is selected for the Phoenix, then Harsh will also play for the Phoenix.

Which of these could be a complete and accurate list of players in the Phoenix team?

A.
Gaurav, Kunal, Ojha

B.
Harsh, Manish, Naman, Ojha

C.
Kunal, Harsh, Lalit, Ojha

D.
Lalit, Manish, Naman

Independently solved answer: D

Question 16

Aptitude | Quant Math Ability 2

A painter is paid x rupees for painting every 10 metres of a wall and y rupees for painting every extra metre. During one week, he painted 10 metres on Monday, 13 metres on Tuesday, 12 metres on Wednesday, 11 metres on Thursday, and 12 metres on Friday. What is his average daily earnings in rupees for the five - day week?

A. x + (8/5)y

B. (5x + 9y)/5

C. 10x + (8/5)y

D. 5x + 8y

Independently solved answer: A

Question 17

Aptitude | Quant Math Ability 2

The difference between A's speed and B’s Speed is 3km/hr. C’s speed is twice as that of A and B’s speed is twice as that of D. the difference between C and D speed is 30km/hr. The speed of B is:

A. 18 km/hr

B. 19 km/hr

C. 21 km/hr

D. 24 km/hr

Independently solved answer: D under intended ordering

Question 18

Aptitude | Quant Math Ability 2

Two dice are thrown simultaneously, What is the probability that the sum of the numbers shown on the two dice is 5?

A. 1/6

B. 1/9

C. 2/27

D. 1/12

Independently solved answer: B

Question 19

Aptitude | Puzzles 2

Which of the following will come in place of the question mark (?) in the following sequence so that it completes the sequence?

$L_{12}QK$, $MS_{10}I$, NUG_{8}, $O_{6}WE$, (?)

A. $P_{4}YC$

B. $PY_{4}C$

C. PYC_{4}

D. $PW_{4}C$

Independently solved answer: B

Question 20

Aptitude | Logical Reasoning 1

Arpita, Purna, Rina, Xola, Sofia and Zain are sitting in a row. Sofia and Zain are in the center. Arpita and Purna are at the ends. Rina is sitting just to the left of Arpita. _____ is to the immediate right of Purna.

Independently solved answer: Xola

Question 21

Analog | Signals and System

Let the input signal be

$x(t)=e^{-2t}u(t)$

where $u(t)$ is the unit step function.
The system has an impulse response:

$h(t)=e^{-t}u(t)$

What is the value of the output $y(t)$ at t=1, where $y(t) = x(t) * h(t)$?

A.
0.367

B.
0.232

C.
0.632

D.
0.500

Independently solved answer: B

Question 22

Analog | Op amps

The op-amp shown below fig.1 is an ideal op-amp having a finite gain of A_0 is modeled as a HPF(High Pass Filter) shown in fig. 2. The condition of C_X & R_X for which the poles of fig. 1 & fig. 2 will concide will be given as _____

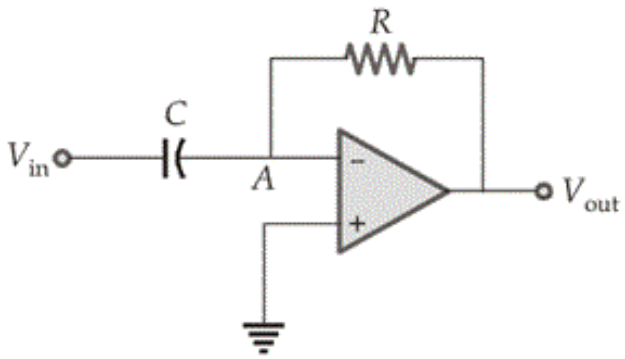


Figure (1)

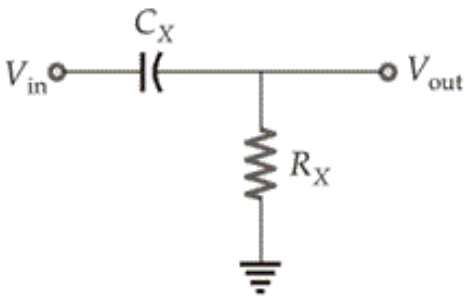


Figure (2)

A.

$C_X = A_0C_1$ & $R_X = R_1/A_0$

B.

$C_X = (1 + A_0)C_1$ & $R_X = R_1/A_0$

C.

$C_X = C_1 / (1 + A_0)$ & $R_X = R_1$

D.

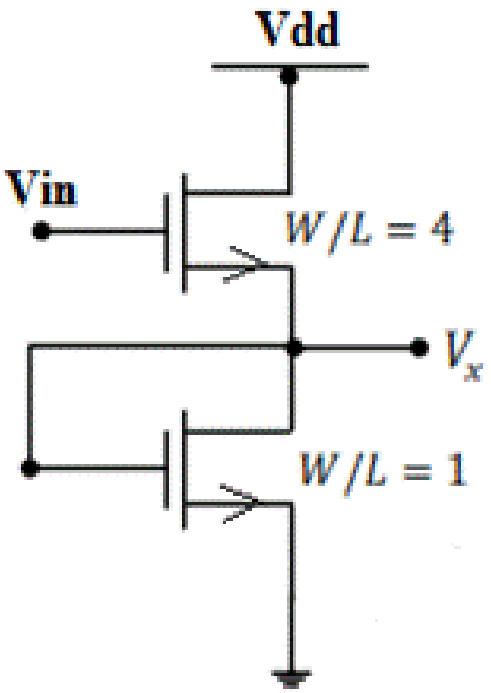
$C_X = C_1 / A_0$ & $R_X = (1 + A_0)R_1$

Independently solved answer: C

Question 23

Analog | MOS and Amplifier

The N-channel MOS transitor has u_nC_{ox}=100uA/V^2 & threshold voltage 1V, for the circuit shown below the value of Vx when Vdd & Vin are set to 6V & 2V respectively will be _____ V



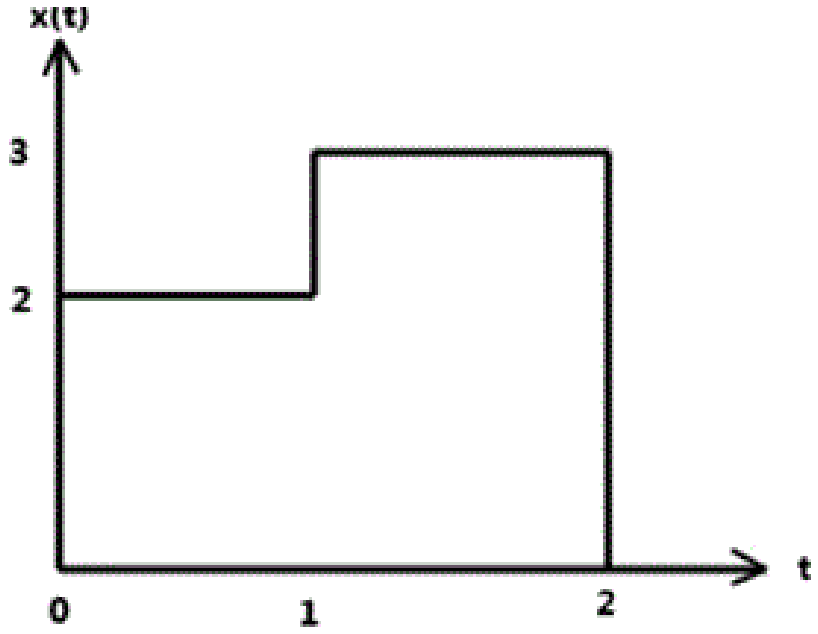
Independently solved answer: 0.5 V

Question 24

Analog | Signals and System

If the Fourier transform of signal x(t) shown below is given by

$$X(j\omega) = \frac{1}{j\omega}\left[A + e^{-j\omega} - 3e^{-j2\omega}\right]$$



, then the value of A will be:

A.
2

B.
3

C.
5

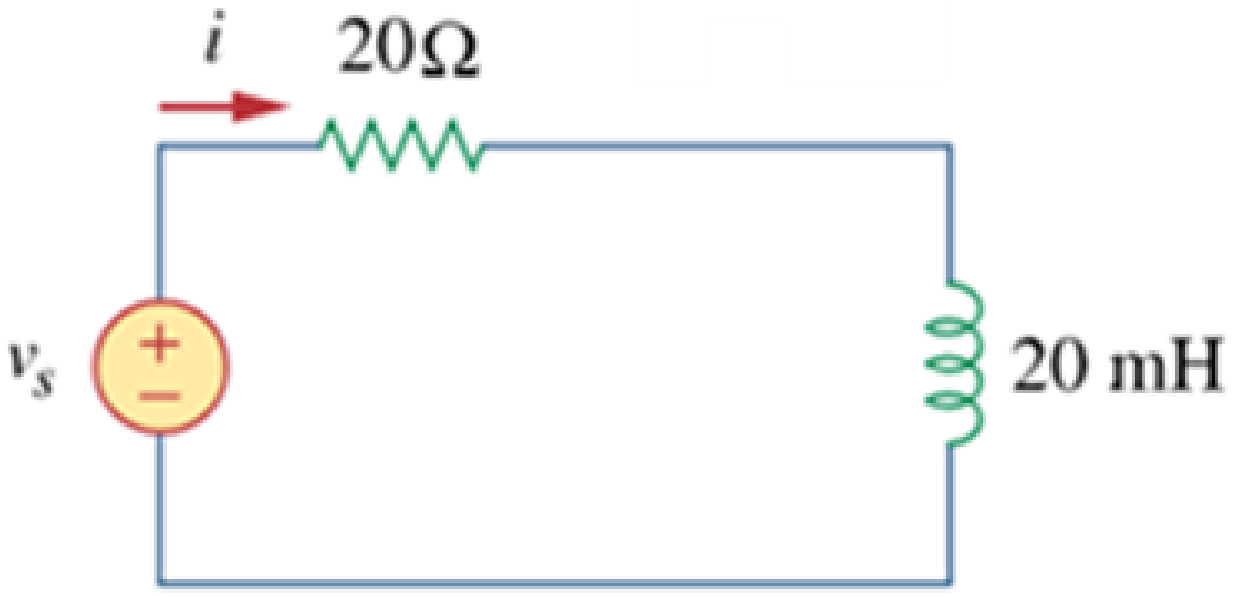
D.
8

Independently solved answer: A

Question 25

Analog | Passive Circuits

In the circuit shown below, if *V(t) = 50 cos 200t V* then the value of current ‘i’ will be:



A.
2.451 cos(200t + 11.31^0) A

B.
2.451 cos(200t - 11.31^0) A

C.
20.4 cos(200t - 11.31^0) A

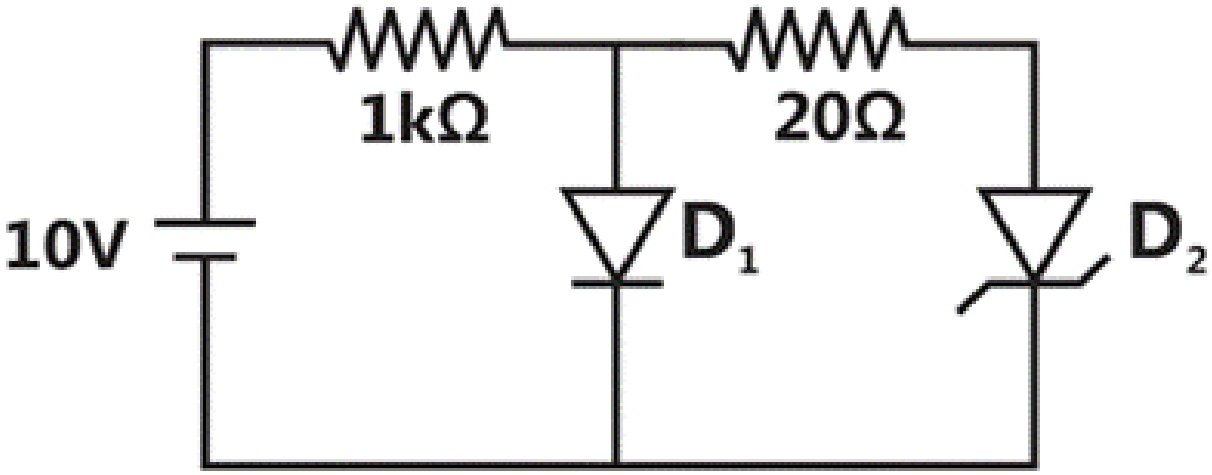
D.
20.4 cos(200t + 11.31^0) A

Independently solved answer: B

Question 26

Analog | Diode and Diode Circuits

In the diode circuit shown assume that the forward voltage drops of the PN diode D1 and Schottky diodeD2 are 0.7 V and 0.3 V, respectively, if ON denotes conducting state of the diode and OFF denotes non conducting state of the diode, then in the circuit the condition will be:



A.
D1 is OFF D2 is ON

B.
D1 is ON D2 is OFF

C.
Both D1 & D2 ON

D.
Both D1 & D2 OFF

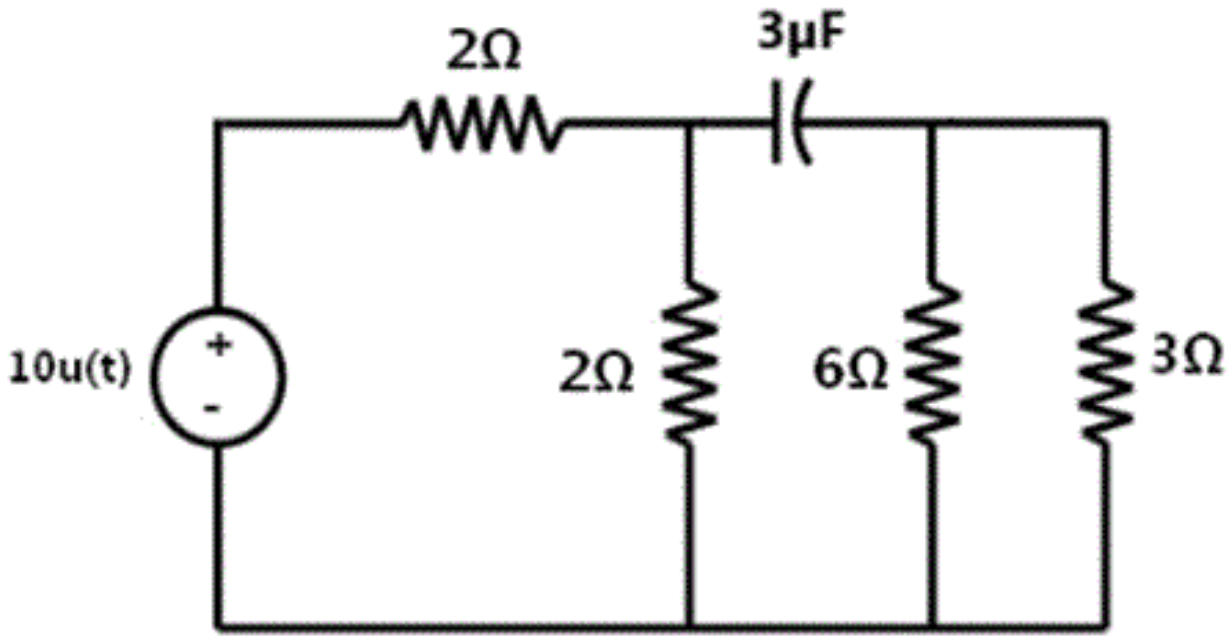
Independently solved answer: A

Question 27

Analog | Passive Circuits

In an industrial motor control panel, a voltage monitoring circuit uses an RC filter to suppress high-frequency transients caused by switching operations. The schematic of the filter section is shown below. The capacitor is rated at 3 μF, and the resistors connected represent the equivalent resistances from different signal paths inside the control logic.

What is the time constant of this RC network that determines how quickly the system responds to a change in input voltage?



A.
3μ sec

B.
9μ sec

C.
15μ sec

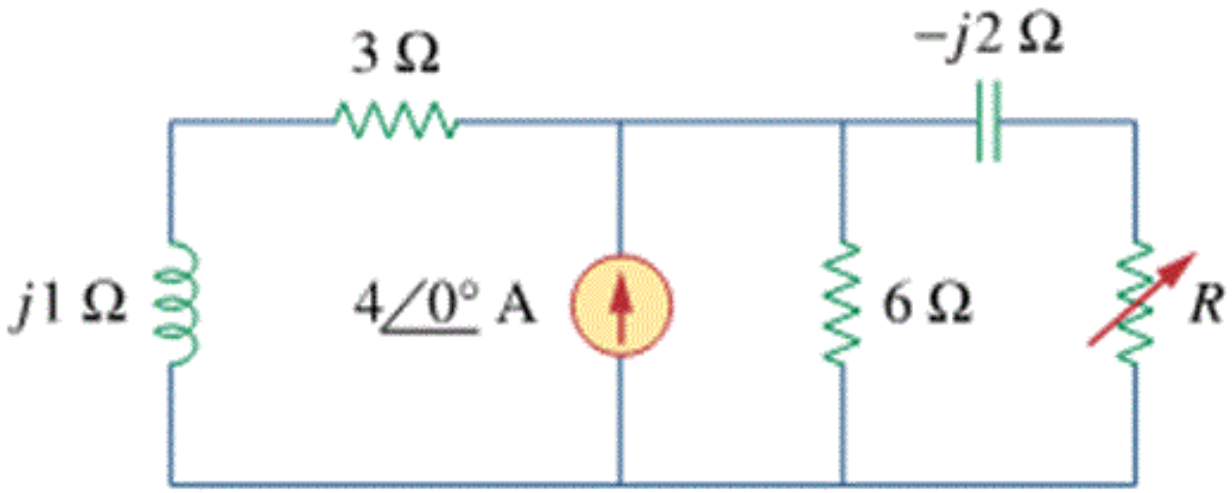
D.
25μ sec

Independently solved answer: B

Question 28

Analog | Passive Circuits

The circuit shown below consists of a variable resistor ‘R’ which is tuned till it absorbs the maximum average power then the maximum average power absorbed will be _____



A.
1.235W

B.
2.451W

C.
3.798W

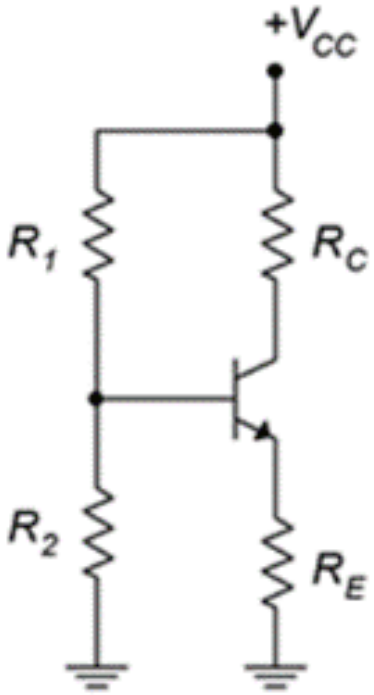
D.
7.285W

Independently solved answer: C

Question 29

Analog | MOS and Amplifier

In NPN amplifier circuit using self bias circuit, Base current I_B = 300 μA and α = 0.98. Find emitter current I_E.



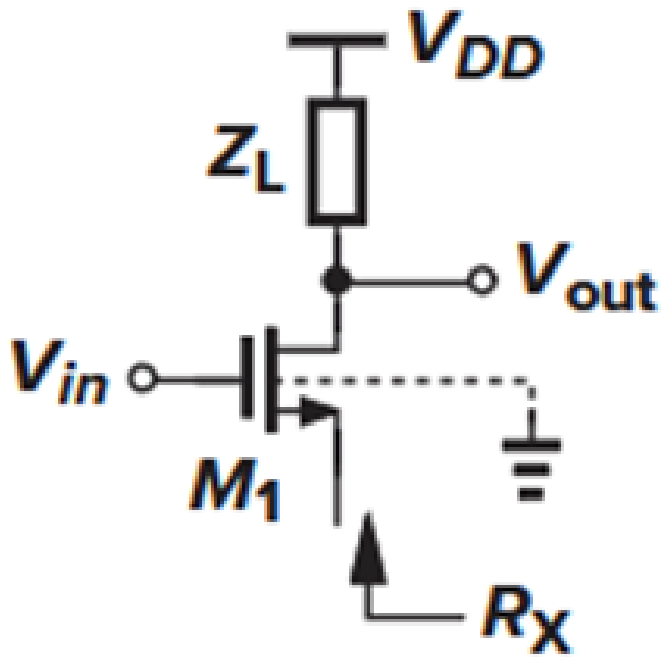
Answer _____ mA

Independently solved answer: 15 mA

Question 30

Analog | BJT MOS

For the MOS circuit shown below, the value of R_x seen through the source terminal will be: _____ (assume λ = 0)



A.

$R_x = 1 / g_{mb}$

B.

$R_x = 1 / g_m$

C.

$R_x = 1 / (g_m + g_{mb})$

D.

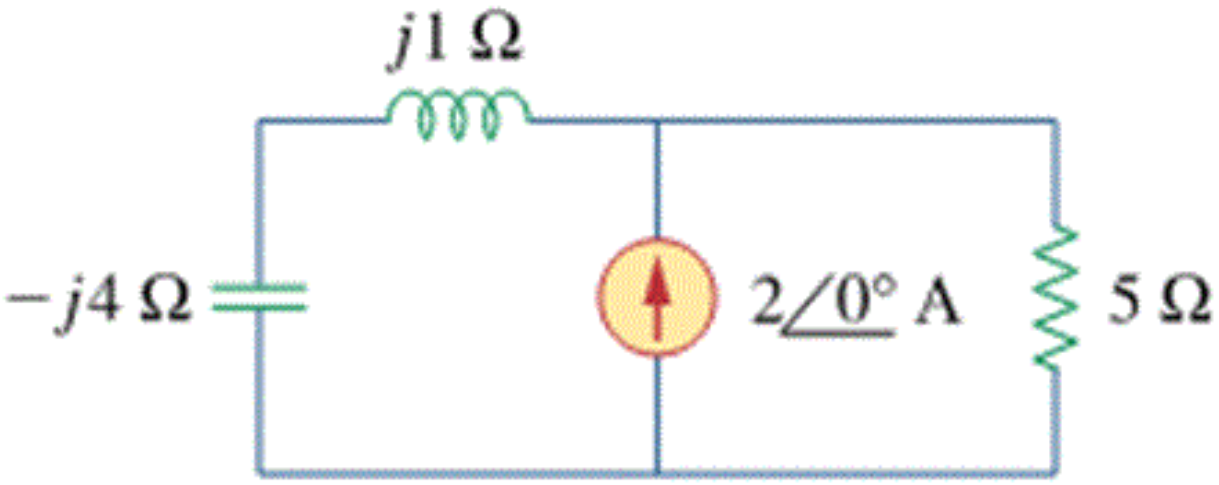
$R_x = g_m / (g_m + g_{mb})$

Independently solved answer: C

Question 31

Analog | Passive Circuits

For the circuit shown below, the average power absorbed by the 5 ohm resistor will be _____



A.
1.21W

B.
5W

C.
3.21W

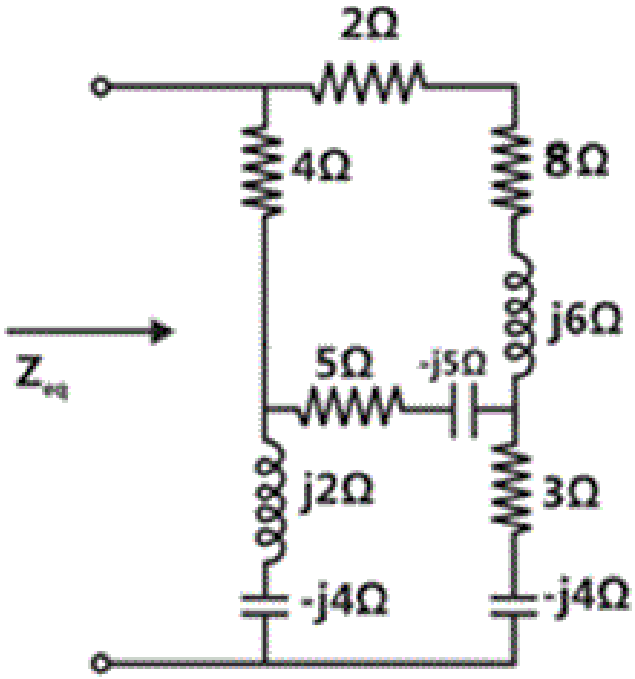
D.
2.64W

Independently solved answer: D

Question 32

Analog | Passive Circuits

The equivalent impedance of a linear passive circuit shown below will be _____



A.

$$\left(\frac{16}{5}+j\frac{14}{15}\right)\Omega$$

B.

$$\left(\frac{14}{15}+j\frac{16}{5}\right)\Omega$$

C.

$$\left(\frac{14}{15} - j\frac{16}{5}\right)\Omega$$

D.

$$\left(\frac{16}{5} - j\frac{14}{15}\right)\Omega$$

Independently solved answer: No exact option

Question 33

Analog | Passive Circuits

A heart rate monitoring device uses an RC circuit to shape the input signal from a skin-contact sensor. The circuit has a resistor of 100 kΩ and a capacitor of 1 µF. When a step input of 5V is applied, the sensor processing unit must wait until the capacitor charges to at least 3.15V before triggering the next stage.

How long must the system wait after the step input is applied to safely trigger the next stage?

A.
23 ms

B.
69 ms

C.
100 ms

D.
50 ms

Independently solved answer: C

Question 34

Analog | BJT MOS

BJT is connected to common emitter mode and operated in active region with collector current Ic =1.01 mA for this when common base current gain is 0.98 and reverse bias saturation current is 0.5 μA then what is the base drive current?

A. 0.2mA

B. 0.2 μA

C. 0.22μA

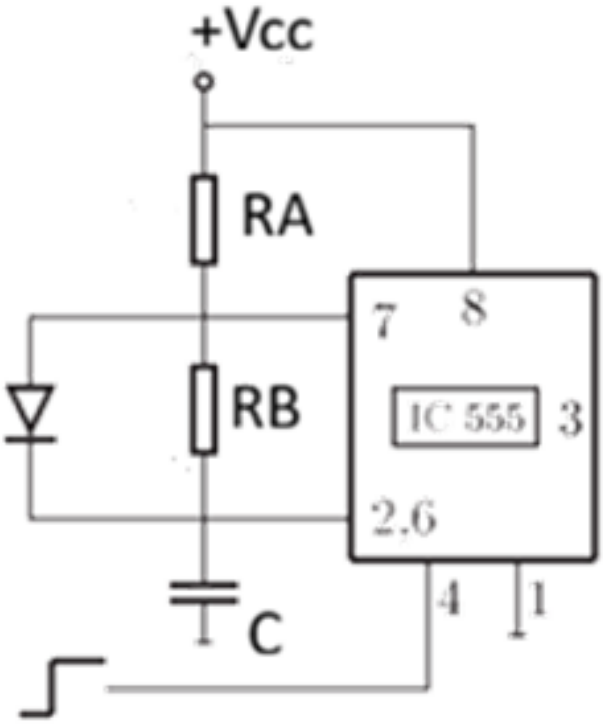
D. 0.3μA

Independently solved answer: No valid option: 20.10 microamps

Question 35

Analog | Linear IC

An IC 555 in an astable multivibrator mode is shown below where RA=RB=20KΩ & C=1pF. The pin description is as 1 and 8-supply; 2-trigger; 4-reset; 6-threshold 7-discharge. The relation between charging & discharging time will be :



A.

$T_{Charging} < T_{Discharging}$

B.

$T_{Charging} > T_{Discharging}$

C.

$T_{Charging} = T_{Discharging}$

D.

$T_{Charging} = 1/T_{Discharging}$

Independently solved answer: C, ideal-diode model

Question 36

Analog | Passive Circuits

A switched-capacitor circuit uses a 1 pF capacitor switched between input and virtual ground at a rate of 250 kHz. What is the effective resistance seen at the input?

A.
4 MΩ

B.
400 kΩ

C.
2 MΩ

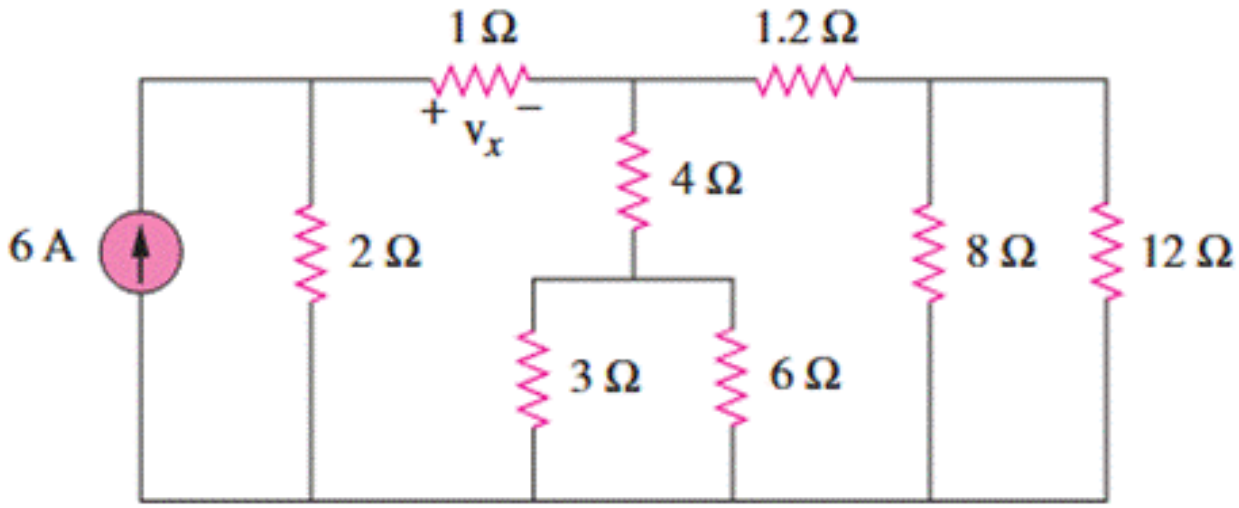
D.
4 kΩ

Independently solved answer: A

Question 37

Analog | Passive Circuits

For the resistive network shown below, the power dissipated by 12Ω resistor will be:



A.
1W

B.
1.5W

C.
1.92W

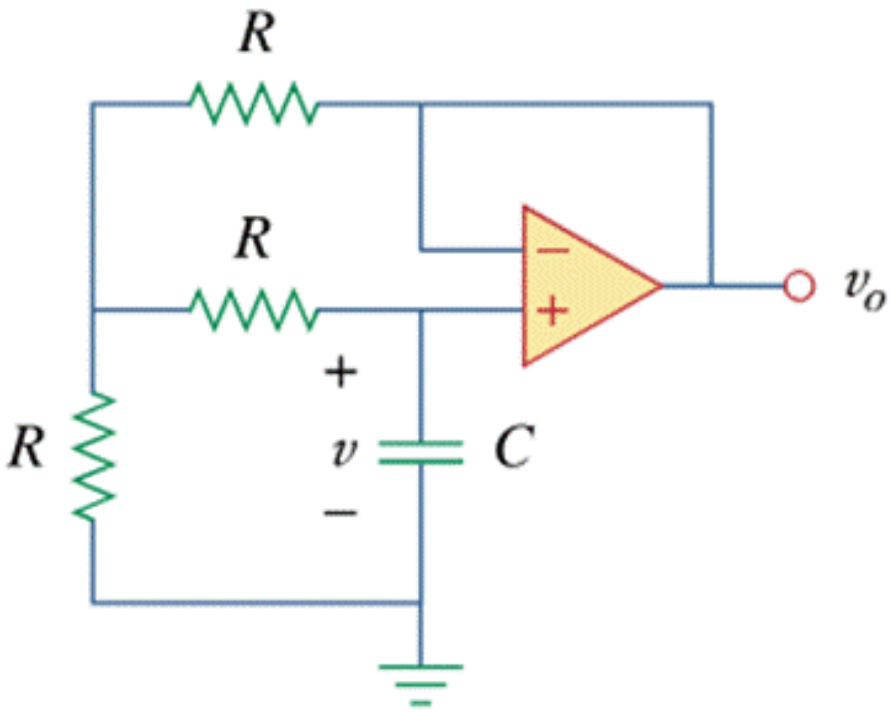
D.
3.84W

Independently solved answer: C

Question 38

Analog | Op amps

The op-amp shown below has initial voltage across capacitor v(0) = 5 V, then for t > 0 the voltage across capacitor v(t) will be _____ (assume R = 10kΩ and C = 1μF)



A.
$e^{-100t/3}\ u(t)\ V$

B.
$2e^{-100t/3}\ u(t)\ V$

C.
$5e^{-100t/3}\ u(t)\ V$

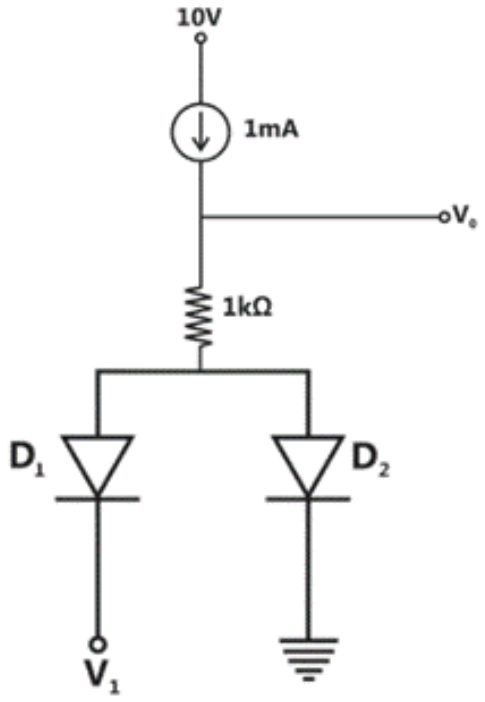
D.
$10e^{-100t/3}\ u(t)\ V$

Independently solved answer: C

Question 39

Analog | Diode and Diode Circuits

For the diode circuit shown below, if V_1 is 1KHz, 5V peak sin wave then the value of $V_{0,peak}$ will be



A.
1 V

B.
-1 V

C.
4 V

D.
-4 V

Independently solved answer: A, ideal diodes

Question 40

Analog | Diode and Diode Circuits

The output voltage across the load resistance of an inductor filter connected to a bridge rectifier shows a DC value of 25 V and a peak-to-peak ripple voltage of 1.5 V. Calculate the ripple factor.

A.
0.03

B.
0.0173

C.
0.0208

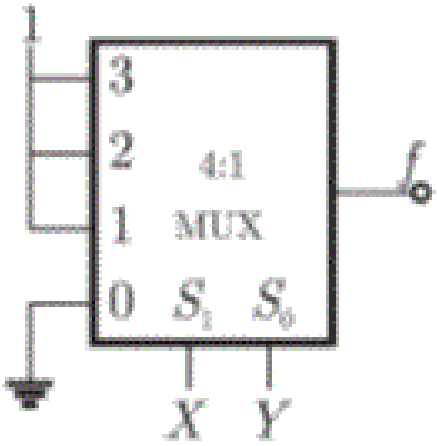
D.
0.0375

Independently solved answer: Waveform-dependent; not A for sinusoidal/triangular ripple

Question 41

Digital | Combinational Logic Design

Find output (f) of the following 4:1 MUX.



A. XY

B. X'Y'

C. X+Y

D. X'+Y'

Independently solved answer: C

Question 42

Digital | Basics of HDL

Consider the following RTL module:
(Code given in verilog and VHDL)

Verilog

```
module seq (
input clk,
input reset,
input sel,
input d,
output reg q
);

always @(posedge clk or posedge reset) begin
if (reset)
q <= 1'b0;
else if (sel)
q <= ~q;
else
q <= d;
end

endmodule
```

VHDL

```
architecture Behavioral of seq is
signal q_reg : STD_LOGIC := '0';
begin
process(clk, reset)
begin
if reset = '1' then
q_reg <= '0';
elsif rising_edge(clk) then
if sel = '1' then
q_reg <= not q_reg;
else
q_reg <= d;
end if;
end if;
```

```
end process;

q <= q_reg;
end Behavioral;
```

Starting from the reset state, the following inputs are applied at four consecutive rising edges:

Edge	sel	d
1	0	1
2	1	0
3	1	1
4	0	0

What is the final value of q?

A.
0

B.
1

C.
X

D.
Z

Independently solved answer: A

Question 43

Digital | Memory Types

You are designing a digital control system for a smart irrigation unit using dual 4-to-1 multiplexers. The system has:

4 soil moisture sensors (8-bit output each)
4 temperature sensors (8-bit output each)
2 control lines to select which sensor pair to read
Both multiplexers share the same control lines

If you implement the selection logic using a PROM that stores the control logic only, what is the minimum PROM size required?

A.
64 bits

B.
128 bits

C.
256 bits

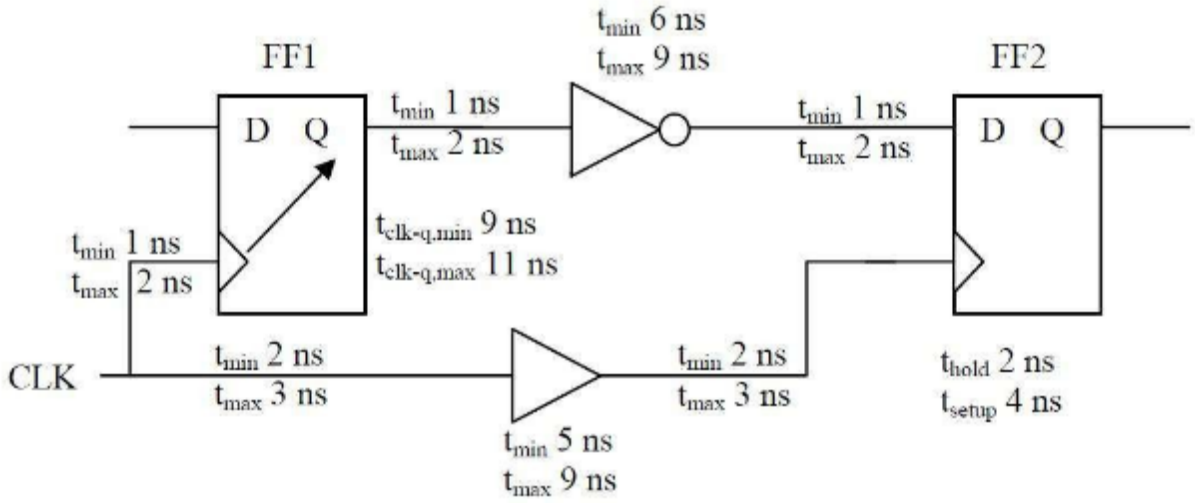
D.
512 bits

Independently solved answer: Inconsistent wording; A likely intended

Question 44

Digital | Static Timing Analysis

For the circuit shown below the set up slack time will be _____ (assume T_{clk} = 10ns)



A. 26ns

B. 21ns

C. 5ns

D. -5ns

Independently solved answer: D under same-corner convention; -11 ns conservative

Question 45

Digital | Basics of C programming

What will be the output of the program given below?

```
#include <stdio.h>

#define $ a

#define _ b

static int $, _;

int main()

{

$ = _ = 2;

printf("%d ", (fun(fun($, _), fun($, _))));

return 0;

}

fun(int $, int _)

{

$ = $ >>= _ <<= 2;

return _;

return $;

}
```

A. 32

B. 0

C. 8

D. Compile time error

Independently solved answer: A=32 likely intended; not portable standard C

Question 46

Digital | Number System

A BCD (Binary Coded Decimal) counter counts from 0000 to 1001 and then resets. If the counter is initially at 0110□BCD□ and receives 7 clock pulses, the final count will be _____ (BCD).

Express your answer as a 4-digit binary number.

Independently solved answer: 0011

Question 47

Digital | Basics of C programming

What will be the output of the program given below?

```
#include <stdio.h>
int main()
{
int N = 1111101;
int a = 2;
int res = 0;
while (N != 0) {
res = res + (N % 10) * a;
N = N / 10;
a = a * 2;
}

printf("%d", res);

}
```

A. 250

B. 227

C. 251

D. Compile time error

Independently solved answer: A

Question 48

Digital | Signals and Systems

A discrete-time system satisfies

$$y[n]=3x[n]-2x[n-1]$$

Which property is true?

A.
Non-linear and time varying

B.
Linear and time invariant

C.
Non-linear but time invariant

D.
Linear but unstable

Independently solved answer: B

Question 49

Digital | Digital Logic Gates

A 3-input NAND gate has the following input combinations applied sequentially: ABC = 000, 001, 010, 011, 100, 101, 110, 111. Count the number of times the output is HIGH (logic 1).
The answer is _____

Independently solved answer: 7

Question 50

Digital | Digital Design Flow

An ASIC targeting 1 GHz has completed place-and-route. The timing report shows a worst negative slack (WNS) of -0.18 ns, while all other paths have positive slack. RTL simulation, synthesis, and logical equivalence checking have already passed. The RTL architecture is frozen, and functional changes are not permitted.

Which action is most appropriate to improve timing with the least impact on functionality?

A.
Re-optimize placement and routing around the failing path while preserving logic functionality.

B.
Relax the clock uncertainty so that setup analysis becomes less pessimistic.

C.
Increase the drive strength uniformly across every standard cell in the design.

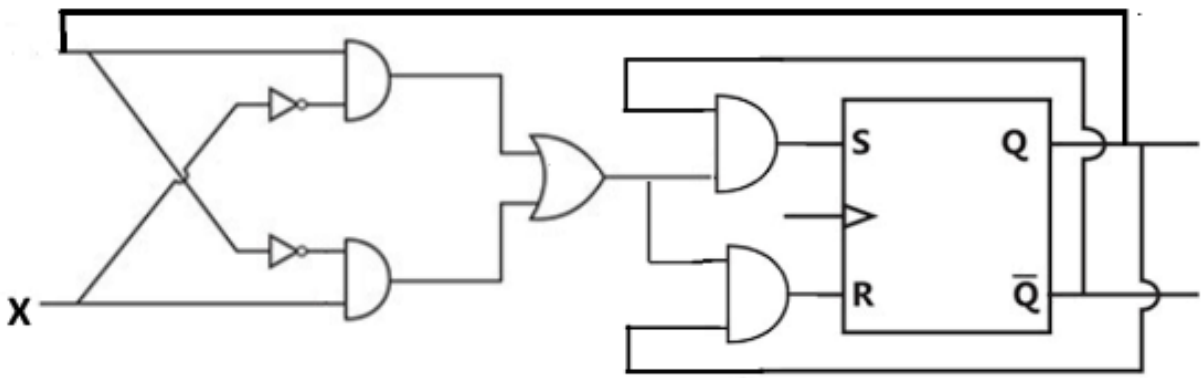
D.
Replace all flip-flops on the failing path with lower setup-time library cells.

Independently solved answer: A

Question 51

Digital | Flipflop

The type of flip flop realized by the circuit shown below will be _____



A. D Flip Flop

B. T Flip Flop

C. JK Flip Flop where J = K = X

D. None of these options

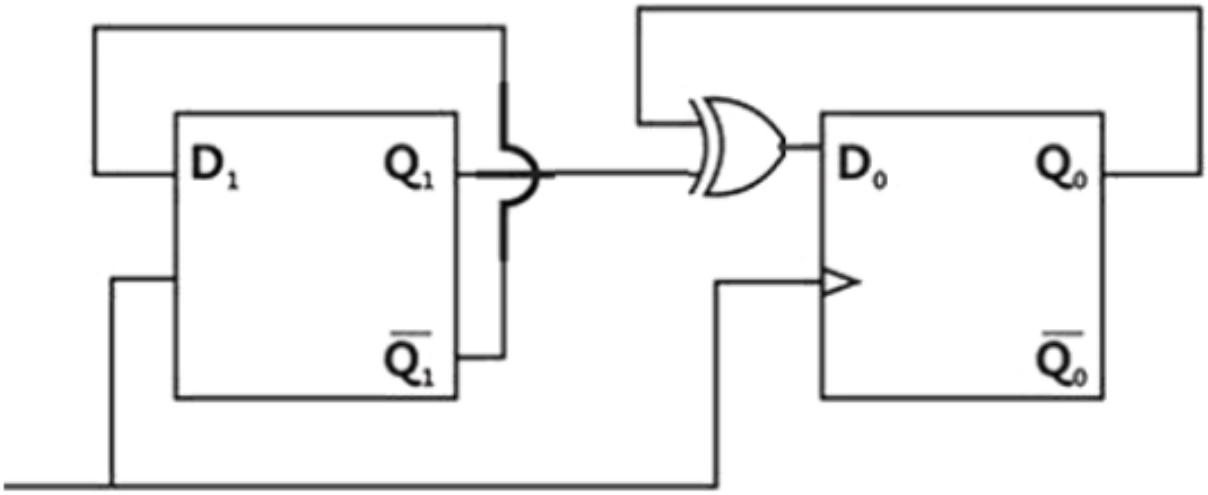
Independently solved answer: A

Question 52

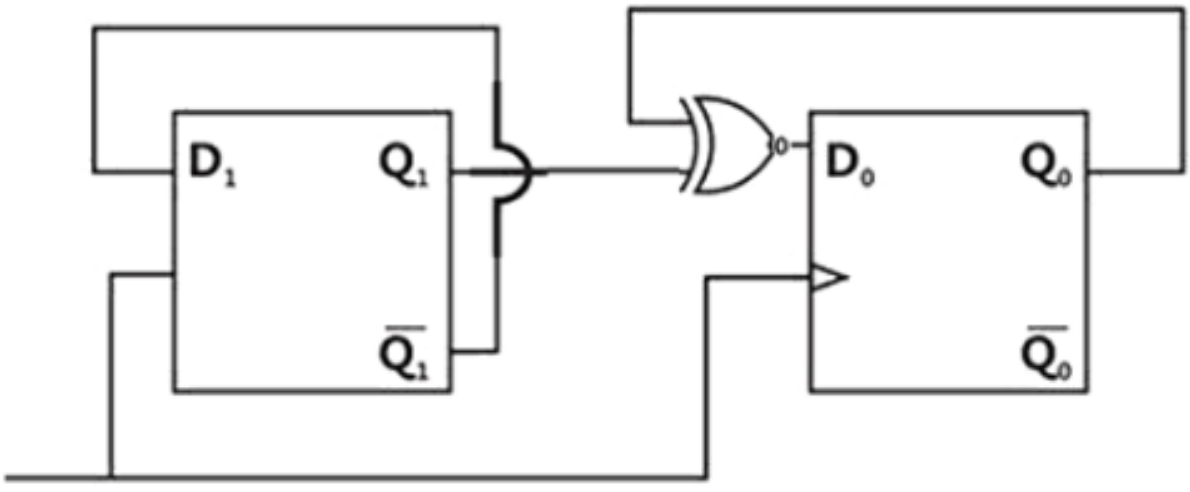
Digital | Sequential Logic design

The correct implementation of a synchronous counter for the count sequence 0 → 3 → 1 → 2 → 0 will be

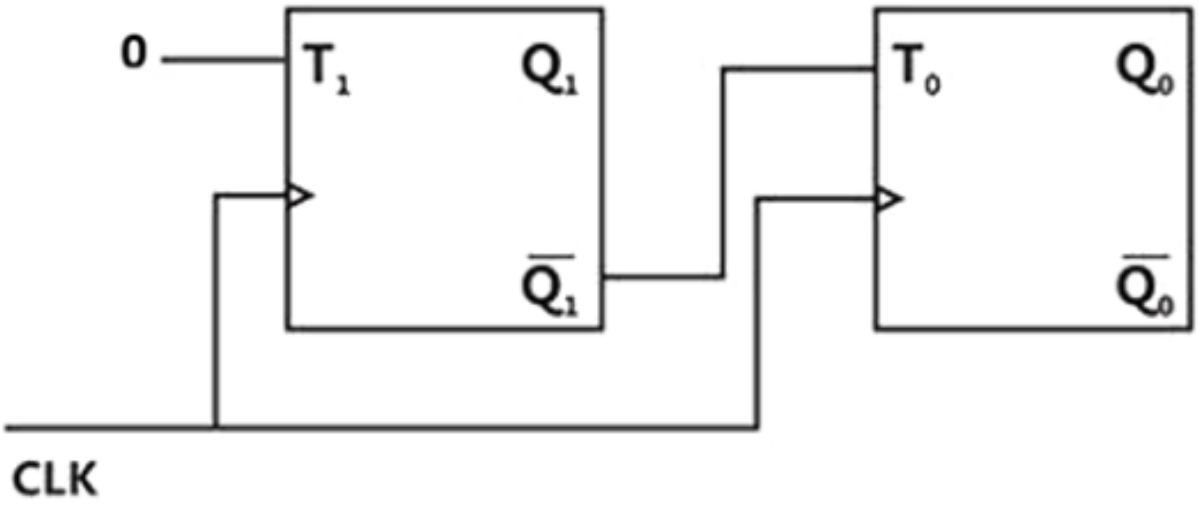
A.



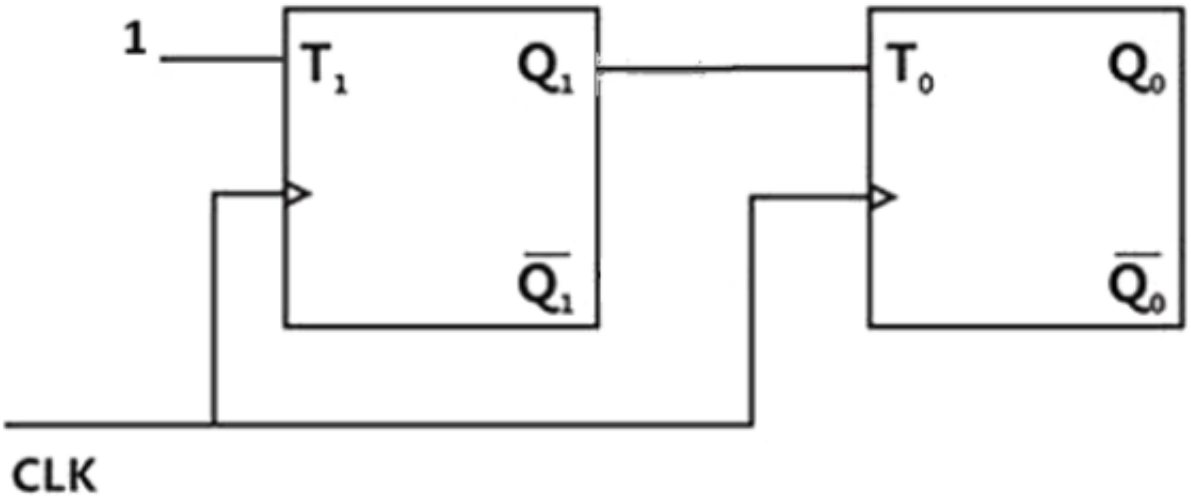
B.



C.



D.



Independently solved answer: B

Question 53

Digital | Memory Types

A traffic light controller system uses a PROM to store the next state logic. The system design requires:

Inputs: 3 state bits + 2 sensor inputs + 1 timer input (total 6 inputs)
Outputs: 3 next state bits + 6 traffic light control bits (total 9 outputs)

Calculate the minimum PROM storage capacity required to implement this sequential logic circuit.

A.
288 bits

B.
576 bits

C.
864 bits

D.
1152 bits

Independently solved answer: B

Question 54

Digital | Microprocessors and Microcontrollers

Consider the following pseudocode executed by a microcontroller:

```
count = 0

while (count < 5)
{
if (INPUT == HIGH)
count = count + 2;
else
count = count + 1;
}
```

If INPUT remains HIGH throughout the execution, what will be the final value of count?

A.
4

B.
5

C.
6

D.
7

Independently solved answer: C

Question 55

Digital | Microprocessors and Microcontrollers

Consider the following 8086 assembly program:

```
MOV AX, 0005H
MOV BX, 0003H
MOV CX, 0000H

L1: ADD CX, AX
DEC BX
JNZ L1

SUB CX, 0004H
```

What will be the final value of the CX register after the program executes?

A.
000BH

B.
000FH

C.
0011H

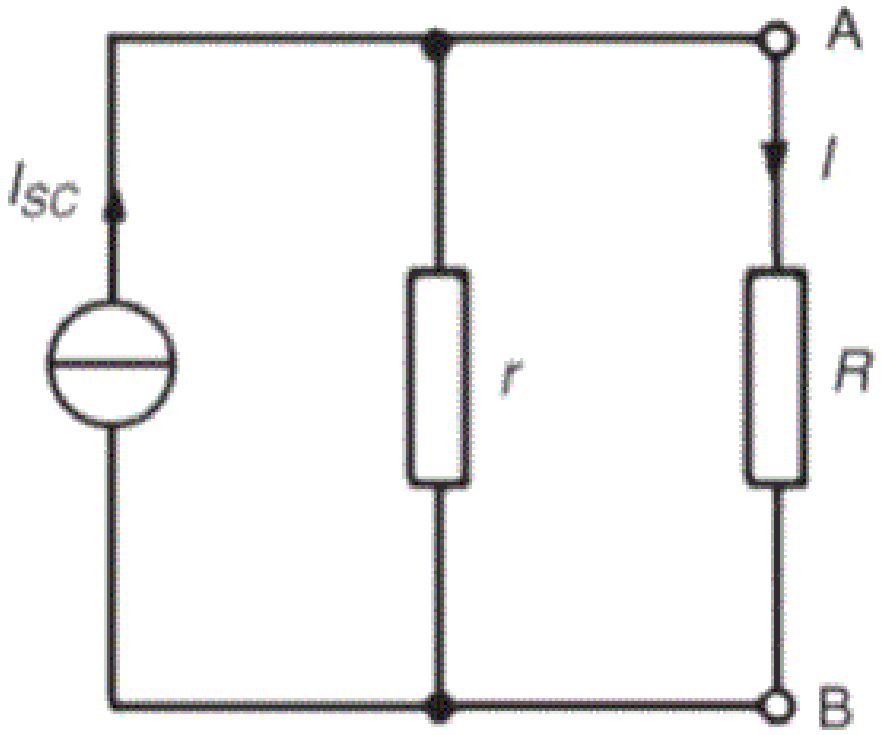
D.
0015H

Independently solved answer: A

Question 56

Digital | Network Theory

If the Norton current and resistance for the circuit shown are respectively are 6A and 4 Ohm, what is the value of $V_{Thevenin}$ (V_{Th}) and $R_{Thevenin}$ (R_{Th})?



A. V_{Th} = 2 V, R_{Th} = 1.5 Ω

B. V_{Th} = 24 V, R_{Th} = 1.5 Ω

C. V_{Th} = 1.5 V, R_{Th} = 1.5 Ω

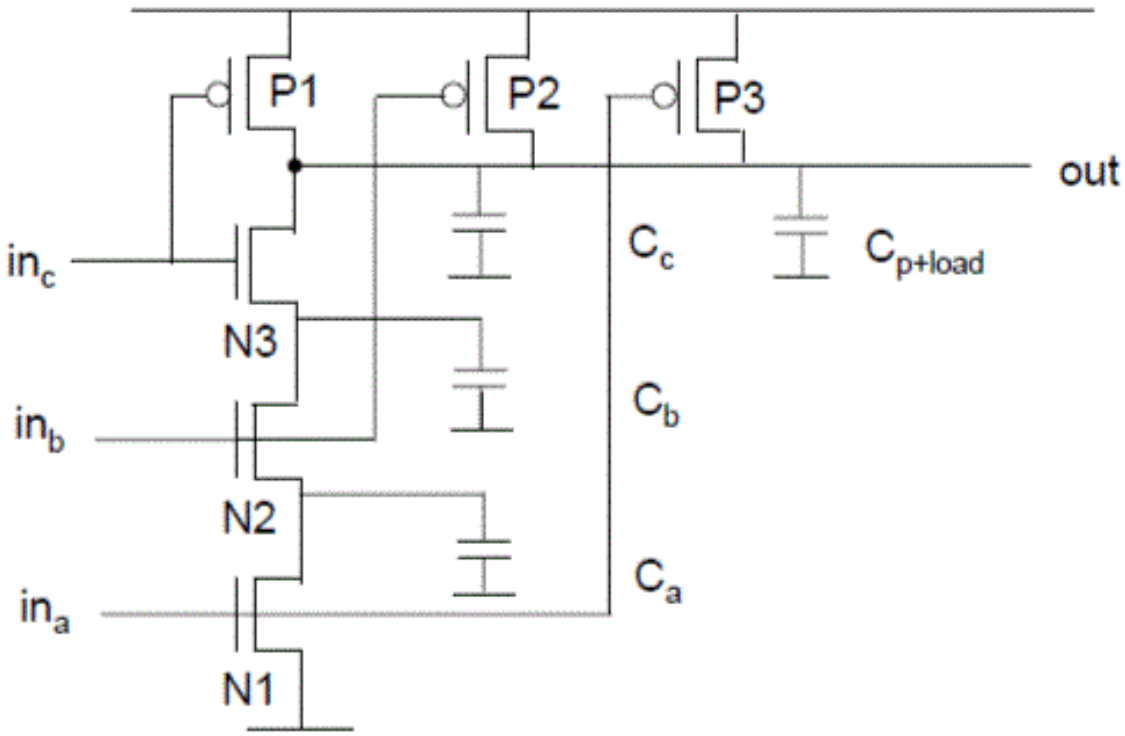
D. V_{Th} = 24 V, R_{Th} = 4 Ω

Independently solved answer: D

Question 57

Digital | CMOS Design

A 3- input NAND gate (NMOS & PMOS based) is showing in figure parasitic elements.



Determine the total time difference of CMOS based NAND gate when the value of Resistance Ra = .5K, Rb = 2K, Rc = 1K, capacitance Ca = 1μF, Cb = 1.5μF, Cc = 0.3μF, and Cpload = .2μF.

A. 6.5msec

B. 10.5msec

C. 16.5msec

D. 9.1msec

Independently solved answer: No matching option under standard RC-tree model

Question 58

Digital | Combinational Logic Design

Consider a four variable based function expressed in terms of its min terms.

F (w, x, y, z) = Σ(0,3, 5, 6, 7,9,10,11,12,14,15)

F (w, x, y, z) can be expressed as (no need to simplify)

A. ∏ (1,2,4,8,13) + (w'x'y'z + w'x'yz' + w'xy'z' + wx'y'z' + wxy'z)

B. Σ(1,2,4,8,13) - ∏ (0,3, 5, 6, 7,9,10,11,12,14,15)

C. Σ(0-15) - ∏ (1,2,4,8,13)

D. ∏ (0-7) + Σ (0-7) - Σ (1,2,4,8,13)

Independently solved answer: D, interpreting subtraction arithmetically

Question 59

Digital | Digital Signal Processing

A sensor signal contains frequency components up to 18 kHz. An engineer samples the signal at 30 kHz without using an anti-aliasing filter.

Which statement is correct

A.
All frequency components are reconstructed correctly.

B.
Frequencies above 15 kHz will alias into the sampled spectrum.

C.
Aliasing occurs only if the sampling frequency is below 18 kHz.

D.
Increasing the ADC resolution eliminates aliasing.

Independently solved answer: B

Question 60

Digital | Counter

A 4-bit ripple counter using JK flip-flops has a propagation delay of 12ns per flip-flop. When the counter transitions from 1111 to 0000, the maximum settling time is _____ ns.

Independently solved answer: 48 ns